

Network Fundamentals

Building a basic network

Wired network connections

Connecting a computer

- RJ45 port
- LAN port



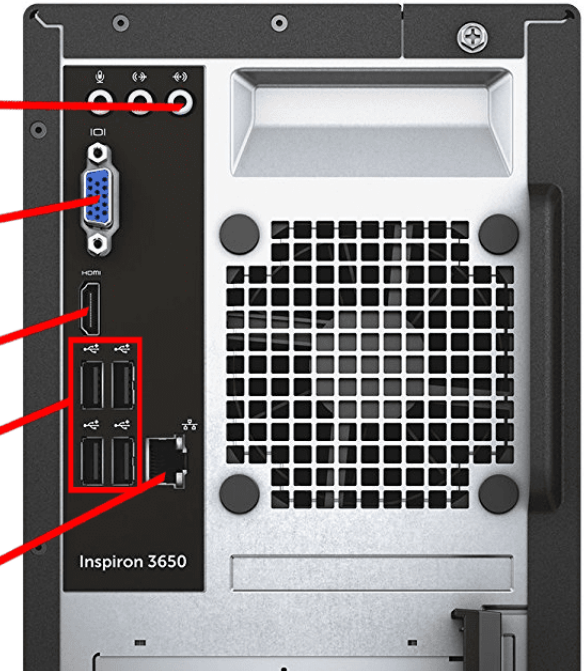
Audio Ports

VGA Port

HDMI Port

USB Ports

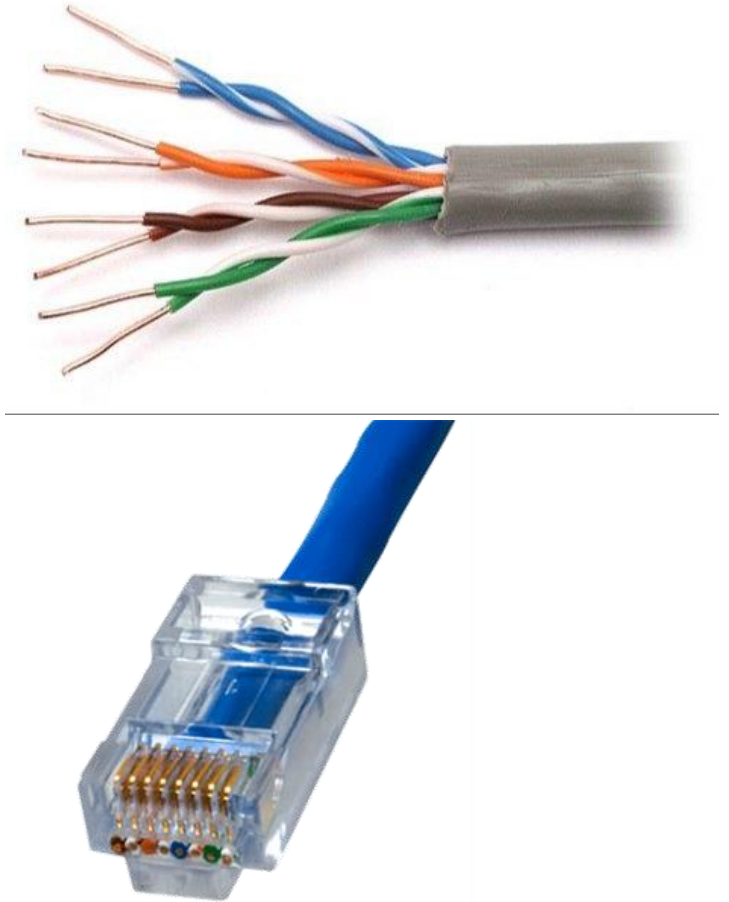
LAN Port



Wired network connections

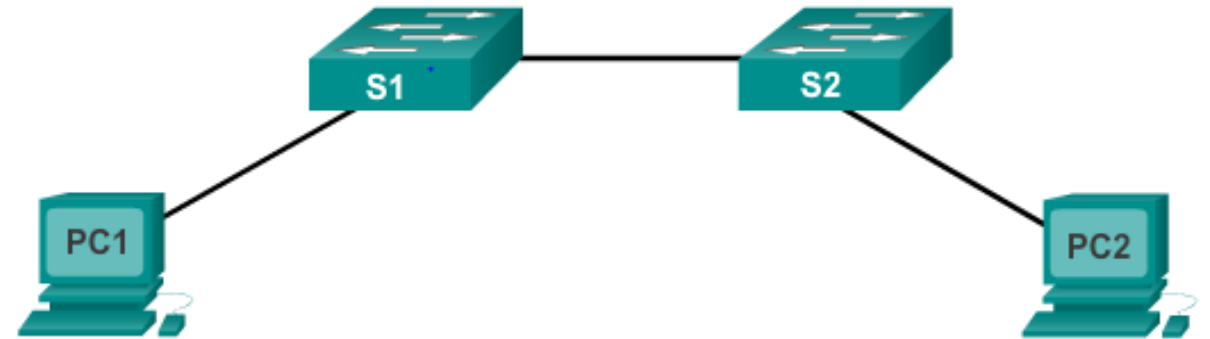
Wires and connectors

- UTP/FTP/STP cables
- Twisted pairs (4 x 2)
- CAT5, CAT5e, CAT6
- RJ45 connector



Switch

- Connecting multiple devices on a local area network
- All devices connected to switch belong to the same **subnet**
- Receive, process, and forward packets



Ip address

- Internet Protocol address
- Logical address for a device in a TCP/IP-based network
- Devices use IP address to communicate over the network
 - Source IP address
 - Destination IP address
- Each end device on a network must be configured with an IP address
 - Static (manual entry)
 - Dynamic (provided by DHCP server)
- Every IP address on a network must be unique

structure of an IPv4 address

Dotted decimal notation

Representing 4 x 8 bits 0.0.0.0 to 255.255.255.255

- 192.168.0.1
- 10.10.5.7
- 240.45.246.233
- ...

IP addresses in private networks

Addresses not routable on the internet
Only for private networks

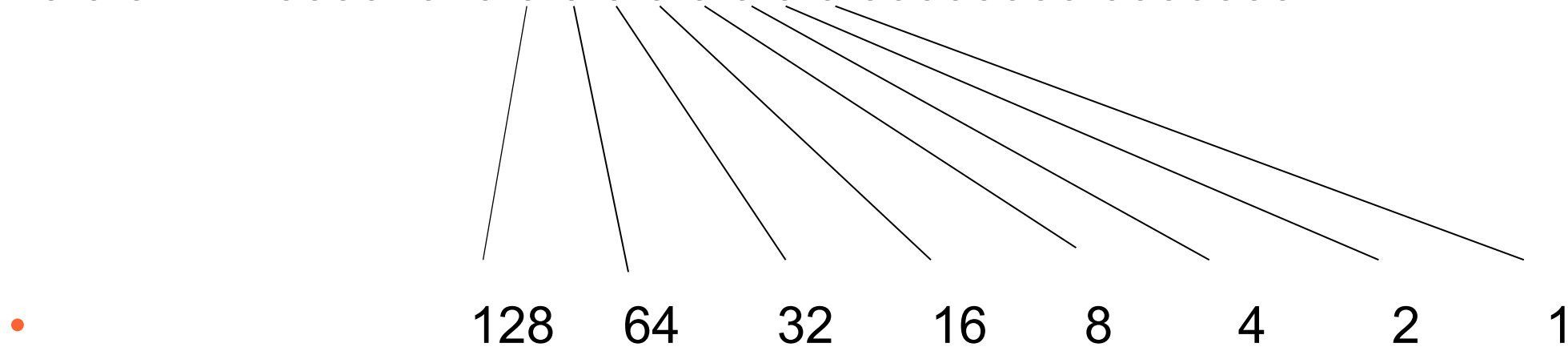
10.0.0.0	– 10.255.255.255
172.16.0.0	– 172.31.255.255
192.168.0.0	– 192.168.255.255

IP Addresses

Dotted decimal represents 4 times 8 bit of binary number

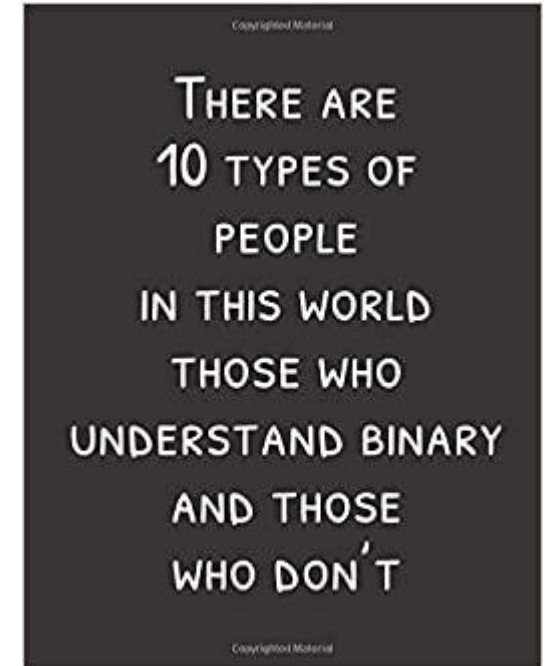
192.168.0.1 → 11000000.10101000.00000000.00000001

10.0.0.1 → 00001010.0 0 0 0 0 0 0 0 0.00000000.00000001



BINARY - DECIMAL

Decimal	Binary
200	
50	
17	
129	
255	
	1110 0001
	1111 1111
	0000 0001
	0010 1111



BINARY - DECIMAL

Decimal	Binary
200	1100 1000
50	0011 0010
17	0001 0001
129	1000 0001
255	1111 1111
225	1110 0001
255	1111 1111
1	0000 0001
47	0010 1111

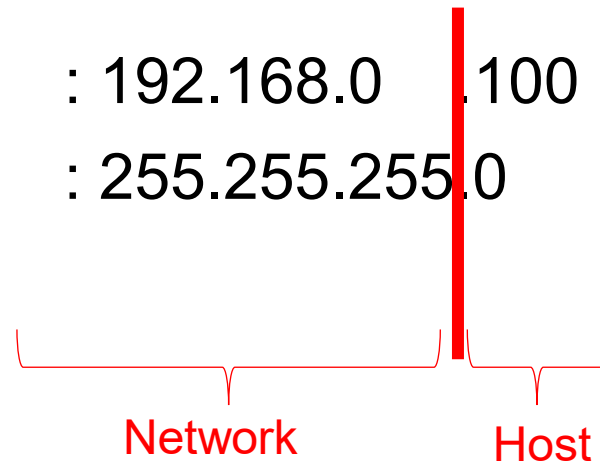
subnetmask

Splits the IP address in a **network part** and a **host part**

- **Network** part needs to be **the same** for all devices in the same subnet
- **Host** part needs to be **unique** for all the devices in the same subnet
- Subnet mask needs to be the same for alle devices in the same subnet

Ip address : 192.168.0 .100

Subnetmask : 255.255.255.0

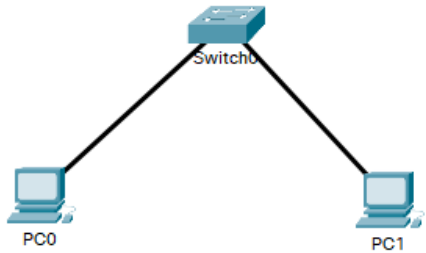


subnetmasks

No unique host part, no unique ip address on the subnet

Different network part, devices don't know they are in the same network

Different subnet mask
→ different network part

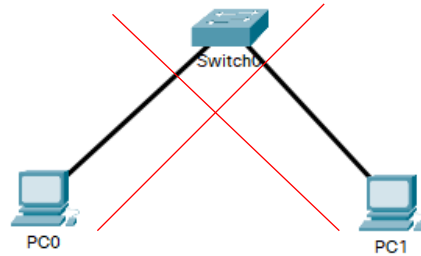


IP: 192.168.0.100
Sn:255.255.255.0

IP: 192.168.0.101
Sn:255.255.255.0

Network: 192.168.0
Host: 100

Network: 192.168.0
Host: 101

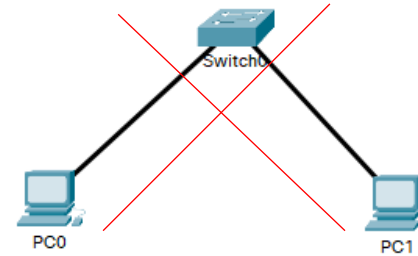


IP: 192.168.0.100
Sn:255.255.255.0

IP: 192.168.0.100
Sn:255.255.255.0

Network: 192.168.0
Host: 100

Network: 192.168.0
Host: 100

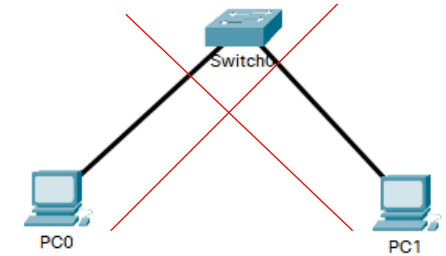


IP: 192.168.0.100
Sn:255.255.255.0

IP: 192.168.2.101
Sn:255.255.255.0

Network: 192.168.0
Host: 100

Network: 192.168.1
Host: 101



IP: 192.168.0.100
Sn:255.255.255.0

IP: 192.168.0.100
Sn:255.255.0.0

Network: 192.168.0
Host: 100

Network: 192.168
Host: 0.100

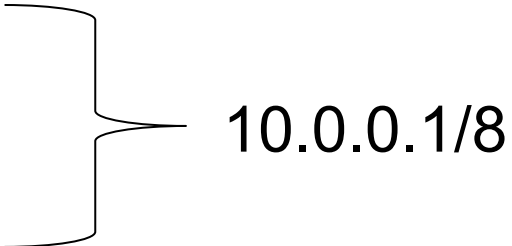
Subnet mask

Subnet mask in dotted decimal notation or CIDR notation

- Binary 1's → network
- Binary 0's → host
- /n gives number of 1's from starting from the left

Subnet mask

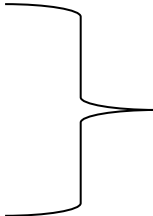
example

- Ip address 10.0.0.1
 - Subnet mask 255.0.0.0
- 
- 10.0.0.1/8

- 00001010.00000000.00000000.00000001 ← ip
 - 11111111.00000000.00000000.00000000 ← subnet = /8
- 
- NETWORK HOST

Subnet mask

example

- Ip address 192.168.0.1
 - Subnet mask 255.255.255.0
- 
- 192.168.0.1/24

- 11000000.10101000.00000000.00000001 ← ip
- 11111111.11111111.11111111.00000000 ← subnet = /24
- NETWORK HOST

Subnet mask

Subnet defines maximum number of devices in a network

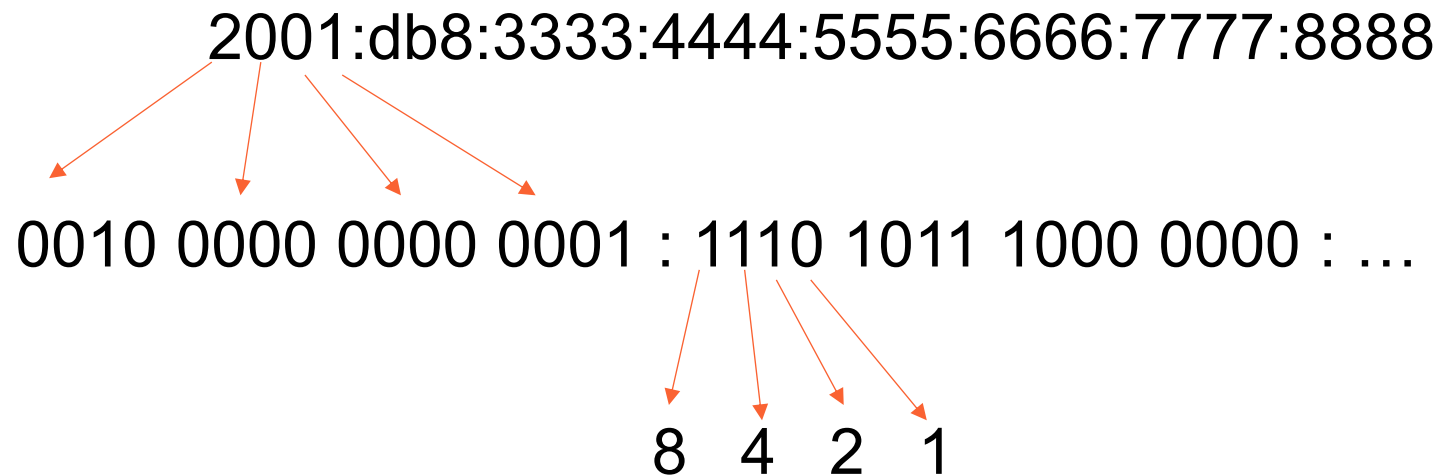
Subnet Mask	CIDR	Binary Notation	Available Addresses Per Subnet
255.255.255.255	/32	11111111.11111111.11111111.11111111	1
255.255.255.254	/31	11111111.11111111.11111111.11111110	2
255.255.255.252	/30	11111111.11111111.11111111.11111100	4
255.255.255.248	/29	11111111.11111111.11111111.11111000	8
255.255.255.240	/28	11111111.11111111.11111111.11110000	16
255.255.255.224	/27	11111111.11111111.11111111.11100000	32
255.255.255.192	/26	11111111.11111111.11111111.11000000	64
255.255.255.128	/25	11111111.11111111.11111111.10000000	128
255.255.255.0	/24	11111111.11111111.11111111.00000000	256
255.255.254.0	/23	11111111.11111111.11111110.00000000	512
255.255.252.0	/22	11111111.11111111.11111100.00000000	1024
255.255.248.0	/21	11111111.11111111.11111000.00000000	2048
255.255.240.0	/20	11111111.11111111.11110000.00000000	4096
255.255.224.0	/19	11111111.11111111.11100000.00000000	8192
255.255.192.0	/18	11111111.11111111.11000000.00000000	16384
255.255.128.0	/17	11111111.11111111.10000000.00000000	32768
255.255.0.0	/16	11111111.11111111.00000000.00000000	65536
255.254.0.0	/15	11111111.11111110.00000000.00000000	131072
255.252.0.0	/14	11111111.11111100.00000000.00000000	262144
255.248.0.0	/13	11111111.11111000.00000000.00000000	524288
255.240.0.0	/12	11111111.11110000.00000000.00000000	1048576
255.224.0.0	/11	11111111.11100000.00000000.00000000	2097152
255.192.0.0	/10	11111111.11000000.00000000.00000000	4194304
255.128.0.0	/9	11111111.10000000.00000000.00000000	8388608
255.0.0.0	/8	11111111.00000000.00000000.00000000	16777216
254.0.0.0	/7	11111110.00000000.00000000.00000000	33554432
252.0.0.0	/6	11111100.00000000.00000000.00000000	67108864
248.0.0.0	/5	11111000.00000000.00000000.00000000	134217728
240.0.0.0	/4	11110000.00000000.00000000.00000000	268435456
224.0.0.0	/3	11100000.00000000.00000000.00000000	536870912
192.0.0.0	/2	11000000.00000000.00000000.00000000	1073741824
128.0.0.0	/1	10000000.00000000.00000000.00000000	2147483648
0.0.0.0	/0	00000000.00000000.00000000.00000000	4294967296

Structure of an IPv6 address

Hexadecimal number grouped in 8 octets

Representing 8 x 16 bits

00:00:00:00:00:00:00:00 to FF:FF:FF:FF:FF:FF:FF:FF

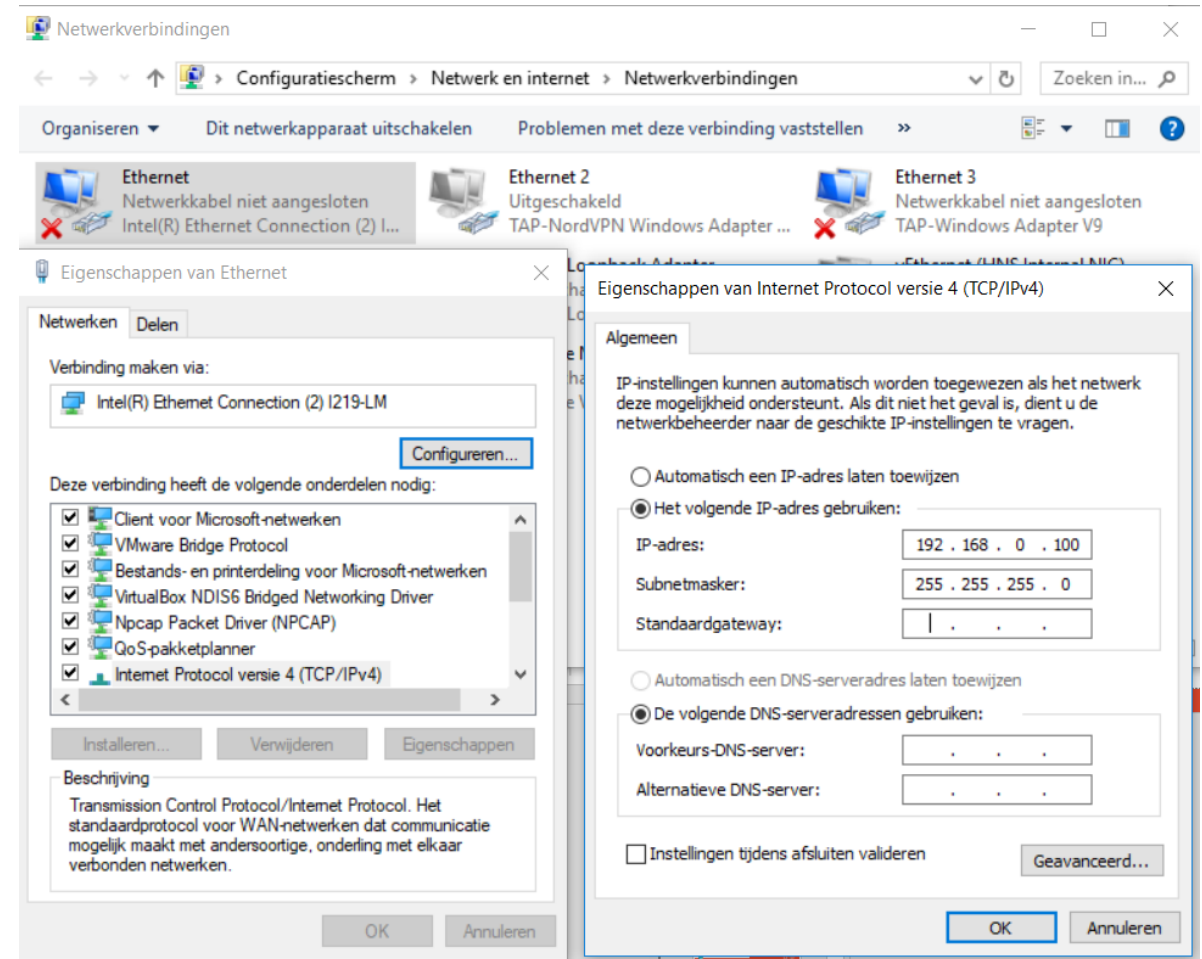


BINARY - hexadecimal

Hexadecimal	Binary
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

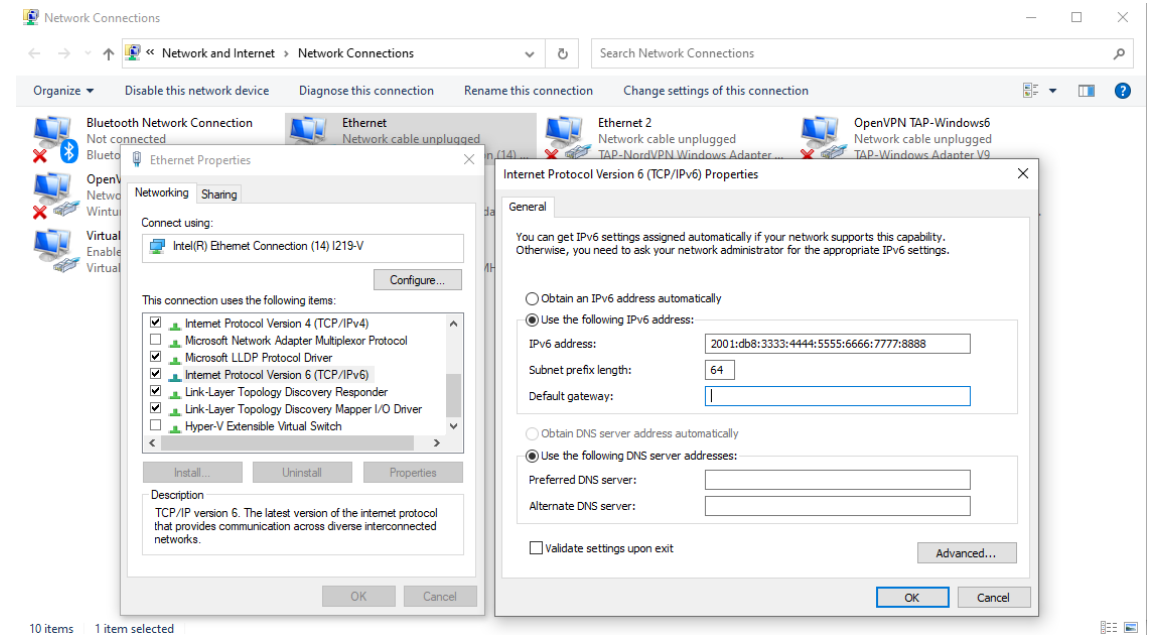
Configuring an IP address

Control panel
Networking center
Change adapter settings
Settings
Internet Protocol v4



Configuring an IPv6 address

Control panel
Networking center
Change adapter settings
Settings
Internet Protocol v6



Automatic IP Address Configuration DHCP

Assigning Dynamic Addresses

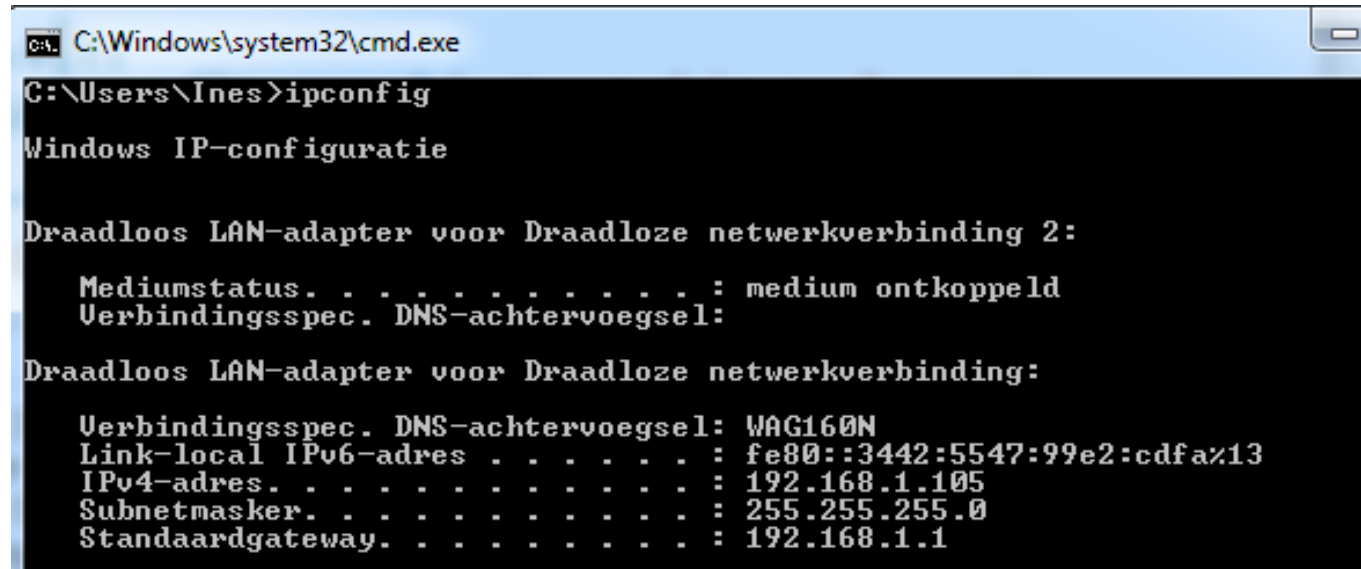
A DHCP server on the network provides devices on the network with an IP address from a pre-configured pool of IP addresses



This property will set the device to obtain an IP address automatically.

Verifying IP Address on windows

ipconfig



```
C:\Windows\system32\cmd.exe
C:\Users\Ines>ipconfig

Windows IP-configuration

Draadloos LAN-adapter voor Draadloze netwerkverbinding 2:

    Mediumstatus. . . . . : medium ontkoppeld
    Verbindingsspec. DNS-achtervoegsel:

Draadloos LAN-adapter voor Draadloze netwerkverbinding:

    Verbindingsspec. DNS-achtervoegsel: WAG160N
    Link-local IPv6-adres . . . . . : fe80::3442:5547:99e2:cdfa%13
    IPv4-adres. . . . . : 192.168.1.105
    Subnetmasker. . . . . : 255.255.255.0
    Standaardgateway. . . . . : 192.168.1.1
```

To see more:

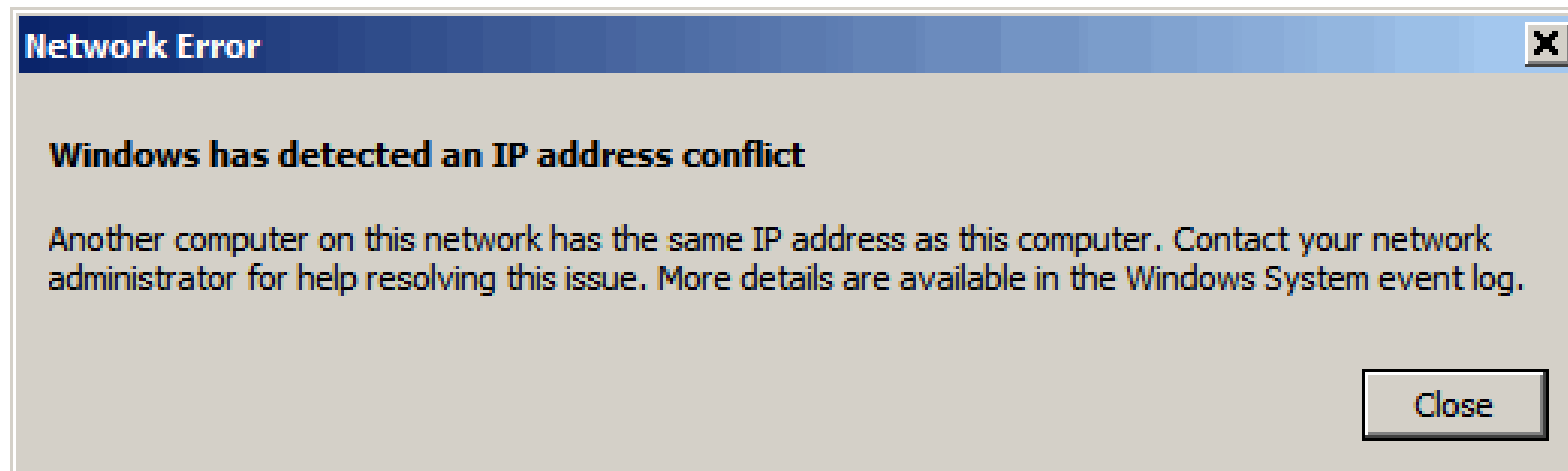
```
C:\Users\Ines>ipconfig /all
```

Verifying ip address on linux and MAC

ip a
(ifconfig)

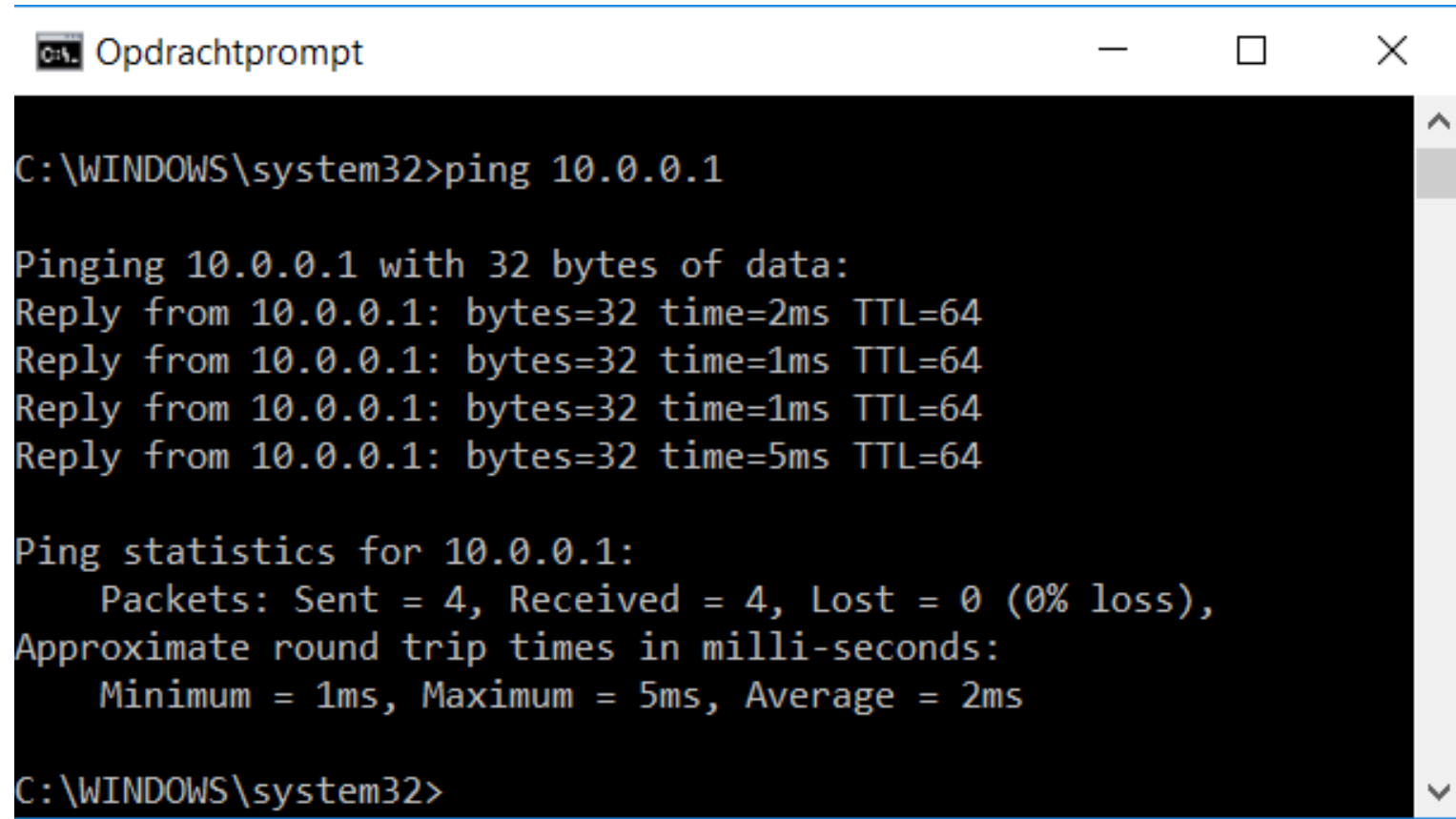
```
ubuntu@ip-10-0-0-200:/var/www/html$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 06:4a:82:4c:59:e7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.200/24 brd 10.0.0.255 scope global dynamic ens5
        valid_lft 2892sec preferred_lft 2892sec
    inet6 fe80::44a:82ff:fe4c:59e7/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ip-10-0-0-200:/var/www/html$
```

IP Address Conflicts



Testing network connectivity

ping



```
C:\WINDOWS\system32>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=2ms TTL=64
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64
Reply from 10.0.0.1: bytes=32 time=5ms TTL=64

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\WINDOWS\system32>
```

Test the Loopback Address on an End Device

127.0.0.1 = localhost = local network port

```
Opdrachtprompt

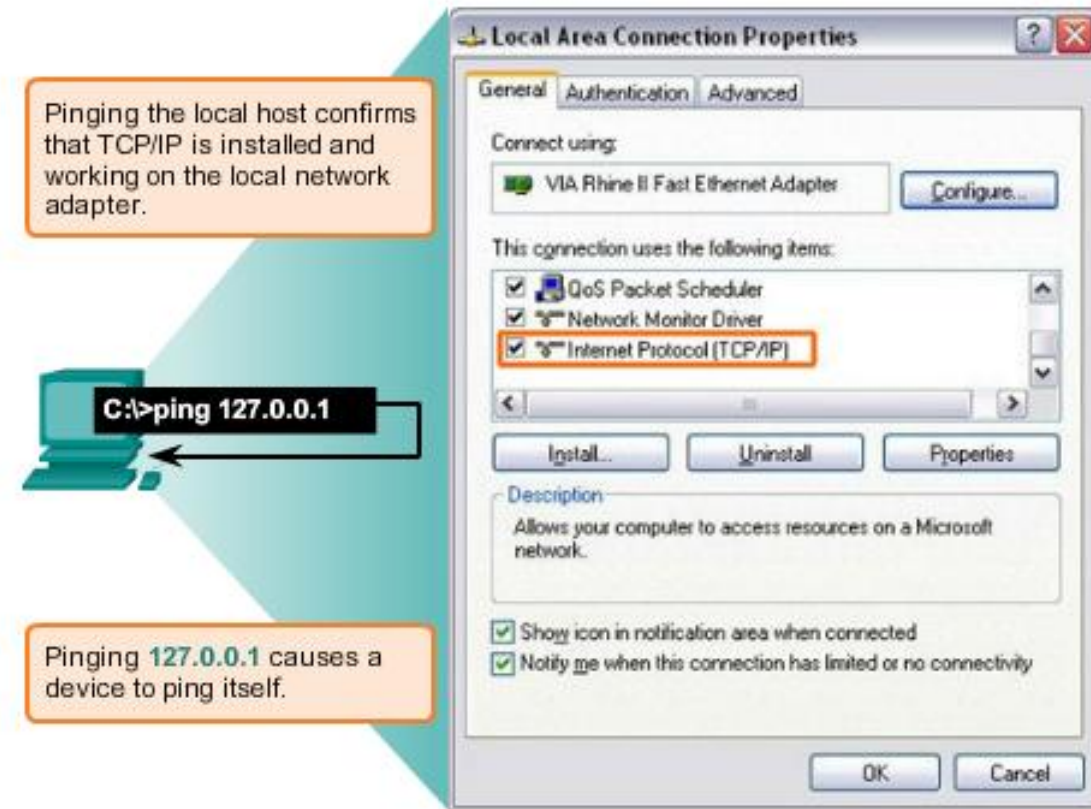
C:\WINDOWS\system32>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

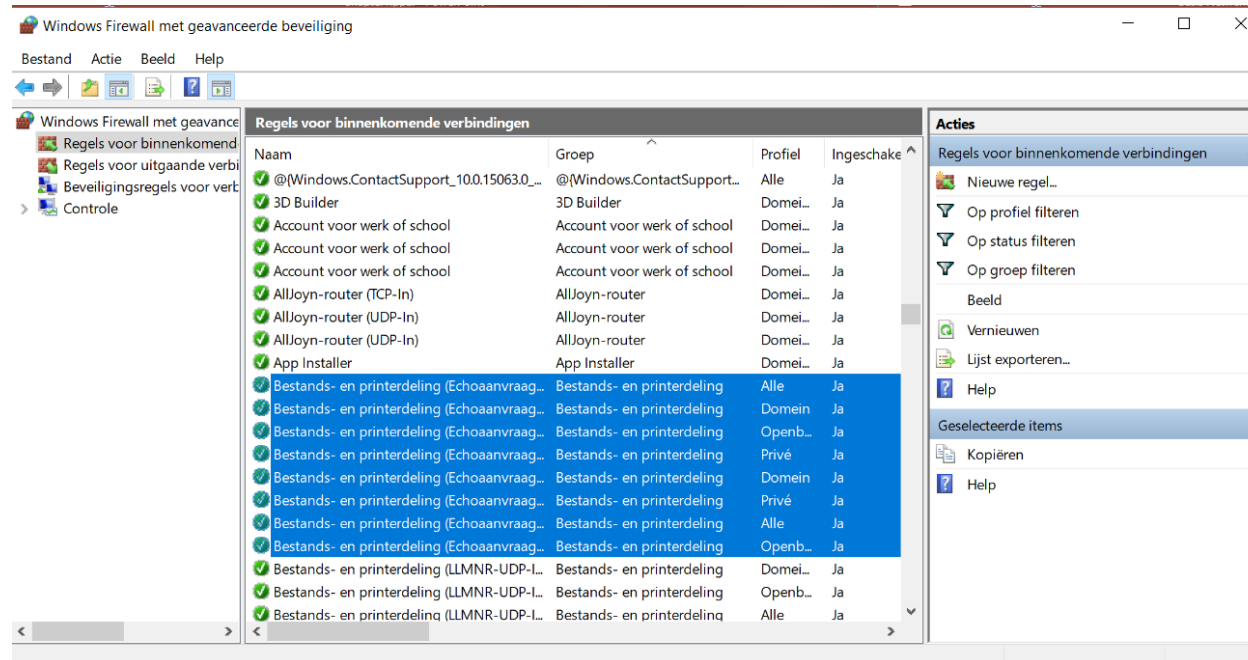
C:\WINDOWS\system32>
```

Testing Local TCP/IP Stack



firewall

Windows blocks ping (icmp protocol) by default
Firewall to block **incoming** traffic

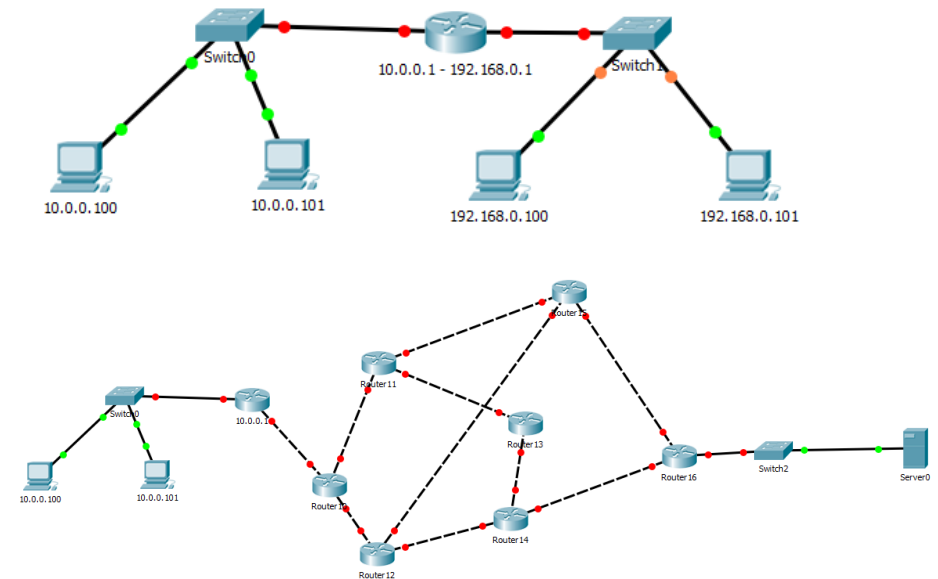


The router

Interconnecting local networks

- Within an organization
- Globally (internet)

Direct/route packets through the network



Default gateway

Get **network** address of destination ip address

Compare to **network** address of localhost ip address

The same?

YES

NO

send to destination

send to default gateway (router)

Algemeen

IP-instellingen kunnen automatisch worden toegewezen als het netwerk deze mogelijkheid ondersteunt. Als dit niet het geval is, dient u de netwerkbeheerder naar de geschikte IP-instellingen te vragen.

☐ Automatisch een IP-adres laten toewijzen

☒ Het volgende IP-adres gebruiken:

IP-adres: 192 . 168 . 0 . 100

Subnetmasker: 255 . 255 . 255 . 0

Standaardgateway: 192 . 168 . 0 . 1

☐ Automatisch een DNS-serveradres laten toewijzen

☒ De volgende DNS-serveradressen gebruiken:

Voorkeurs-DNS-server: . . .

Alternatieve DNS-server: . . .

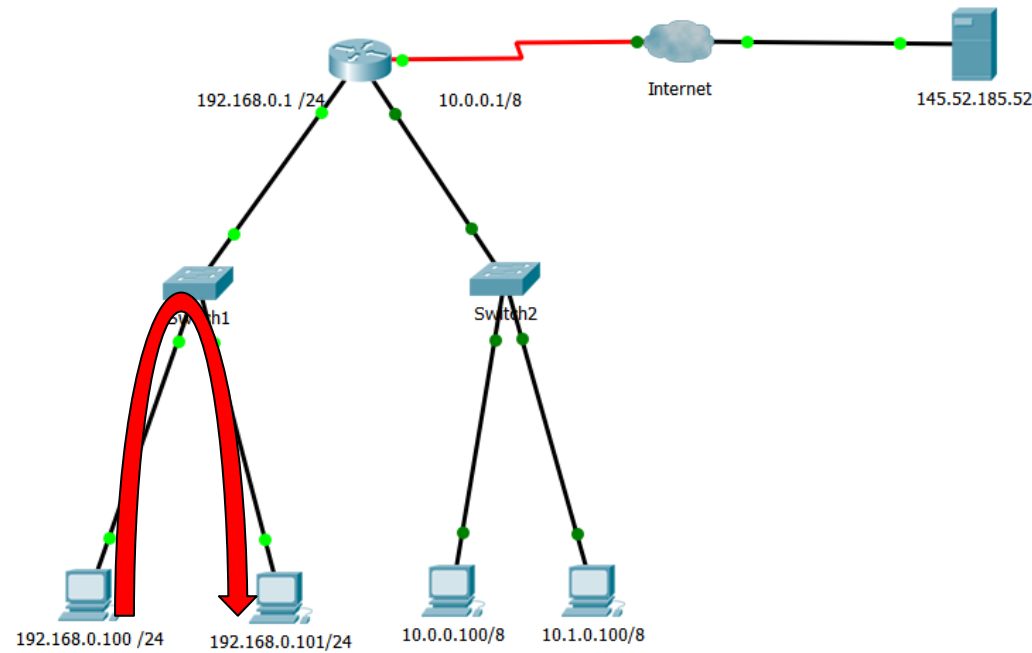
☐ Instellingen tijdens afsluiten valideren

Geavanceerd...

OK Annuleren

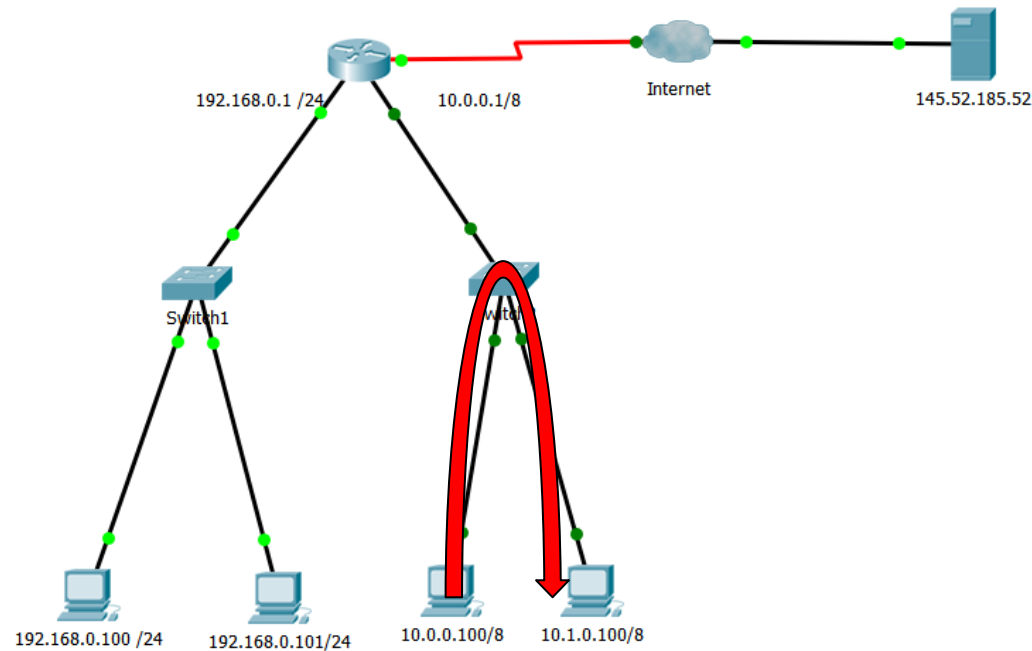
Ip – subnetmask – default gateway

192.168.0.100/24 to 192.168.0.101/24
Same network



Ip – subnetmask – default gateway

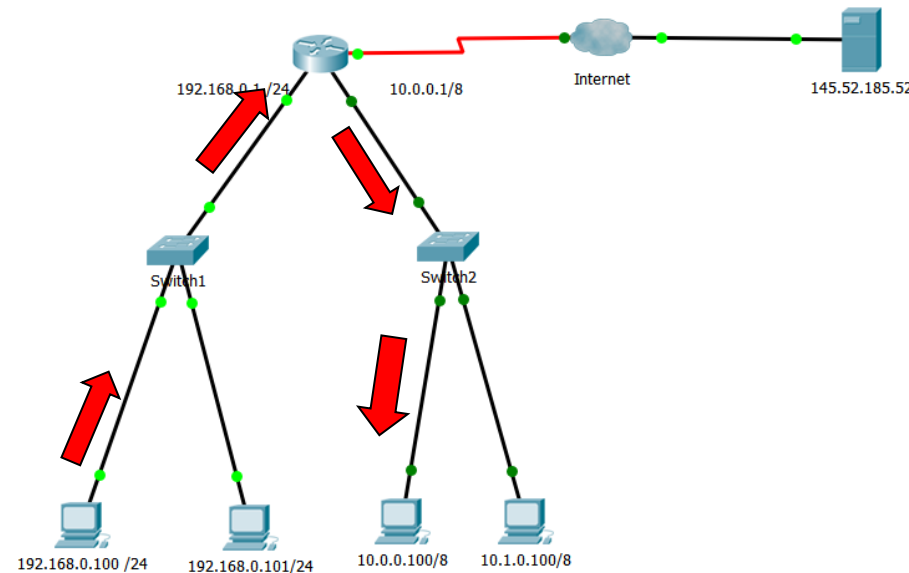
10.0.0.100/8 to 10.1.0.100
Same network



Ip – subnetmask – default gateway

192.168.0.100 to 10.0.0.100/8

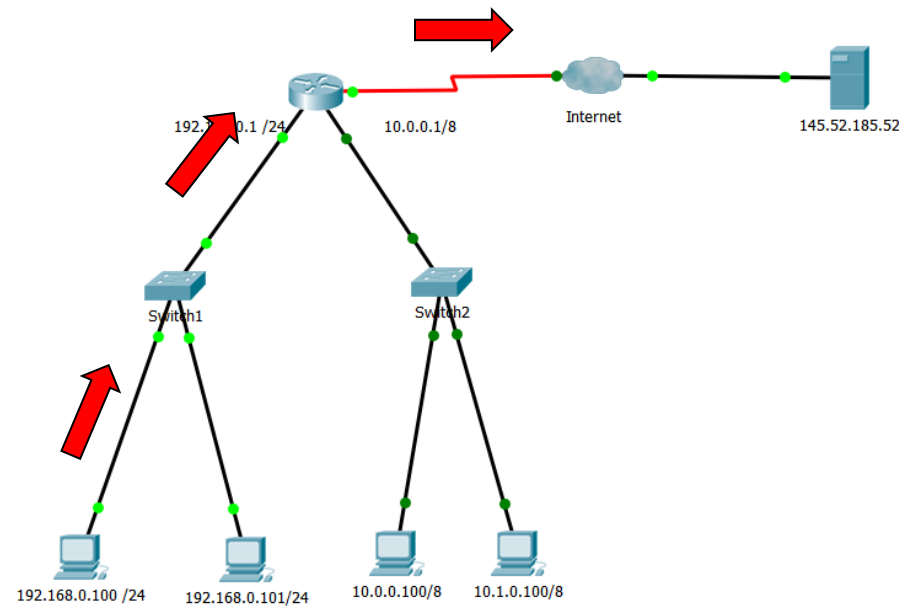
Not same network → through default gateway (router)



Ip – subnetmask – default gateway

192.168.0.100 to webserver

Not same network → through default gateway (router)

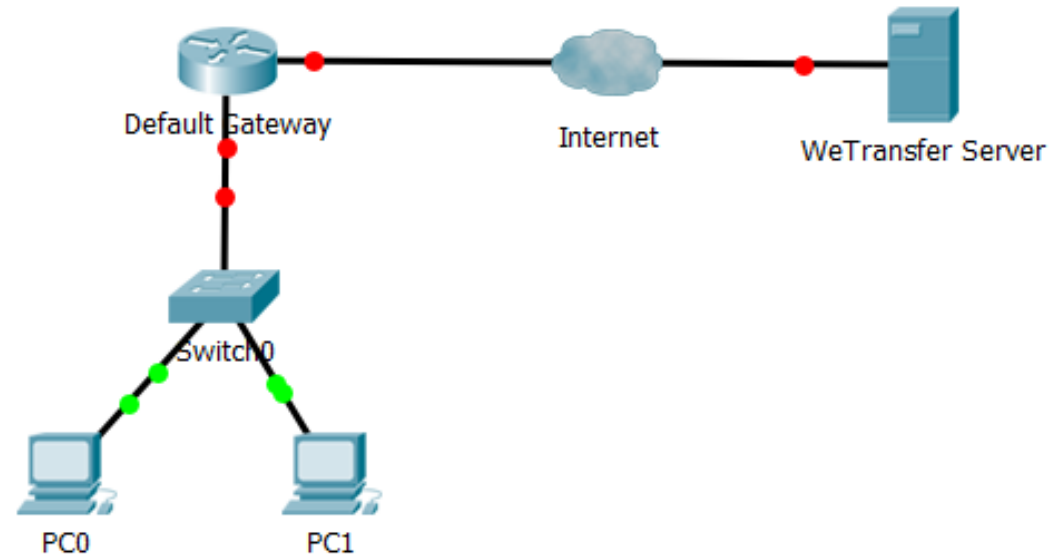
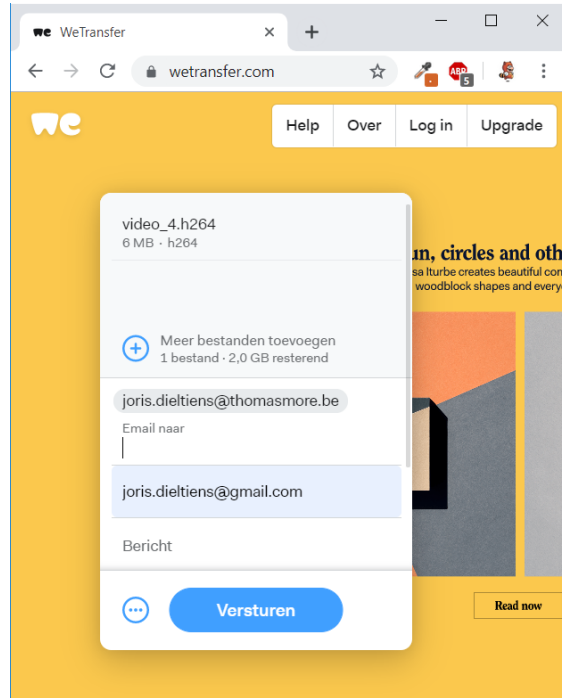


How a packet is send

Very quick overview

Application generates data

- Example: send a large file to WeTransfer server



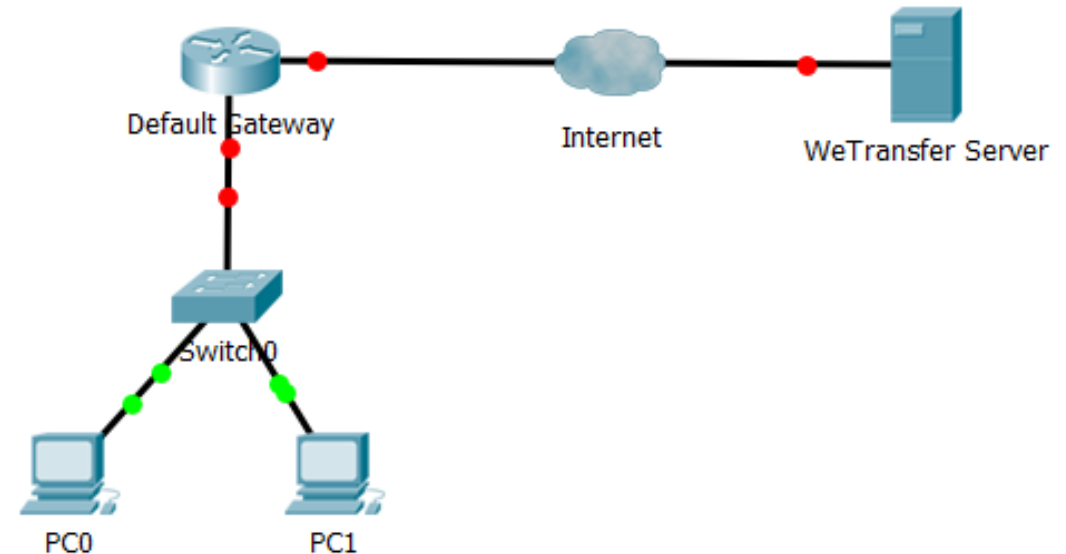
How a packet is send

Very quick overview

Data is split into smaller packets



01101 011001 010011 111010 101011 00111 011111 110110



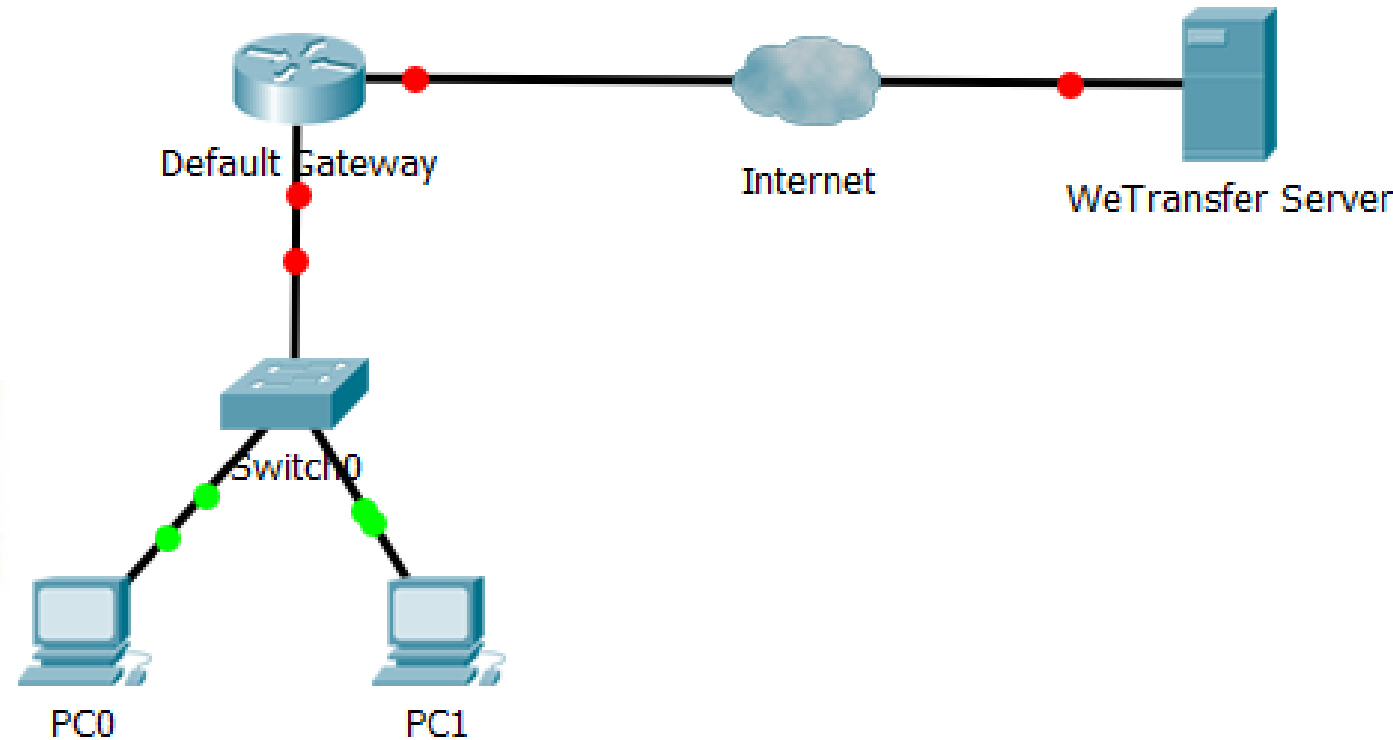
How a packet is send

Very quick overview

Information is added to the packets

- Source ip
- Destination ip
- Sequence number
- ...

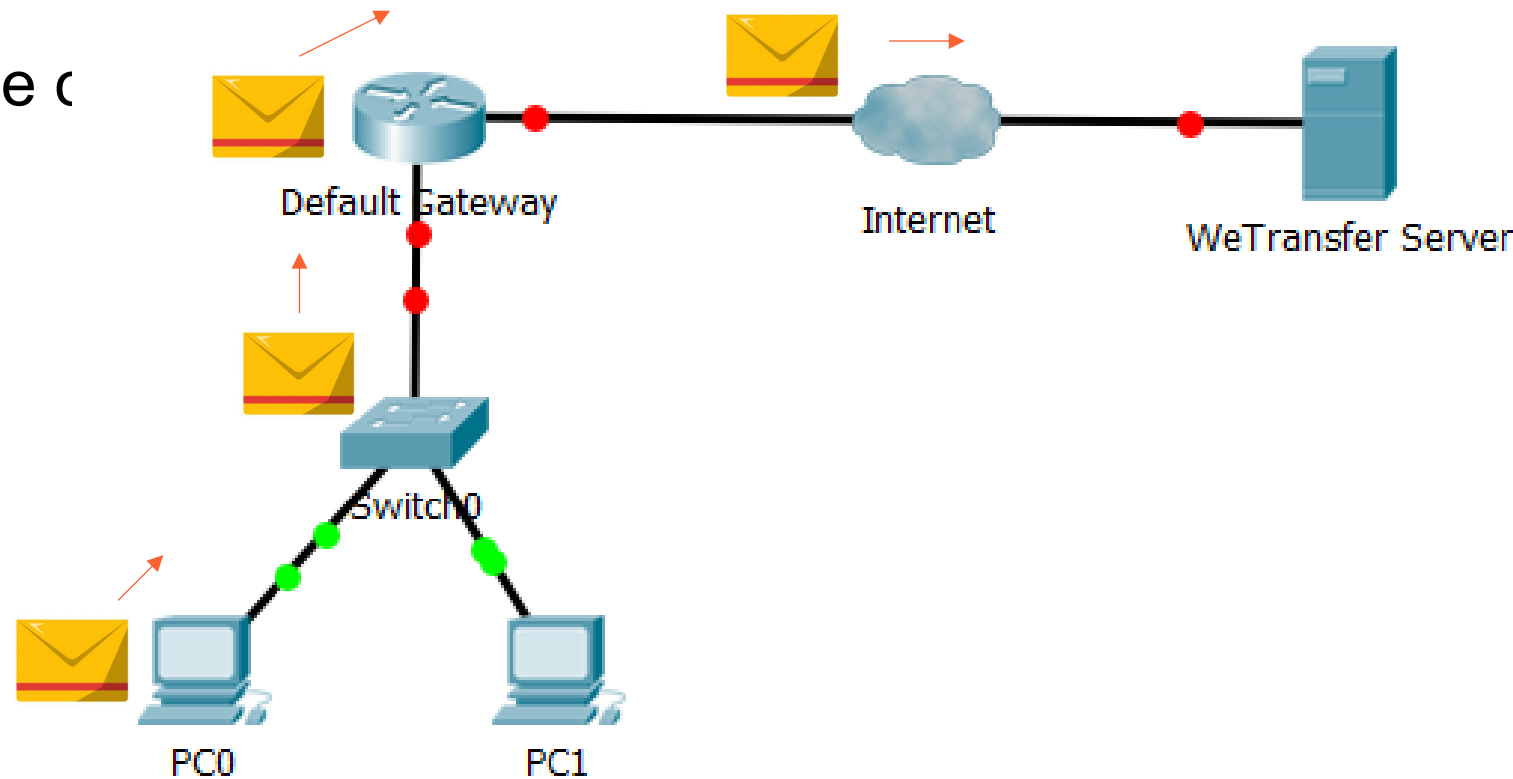
145.51.5.147	10.0.0.1	110110
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How a packet is send

Very quick overview

The packets are send one by one c

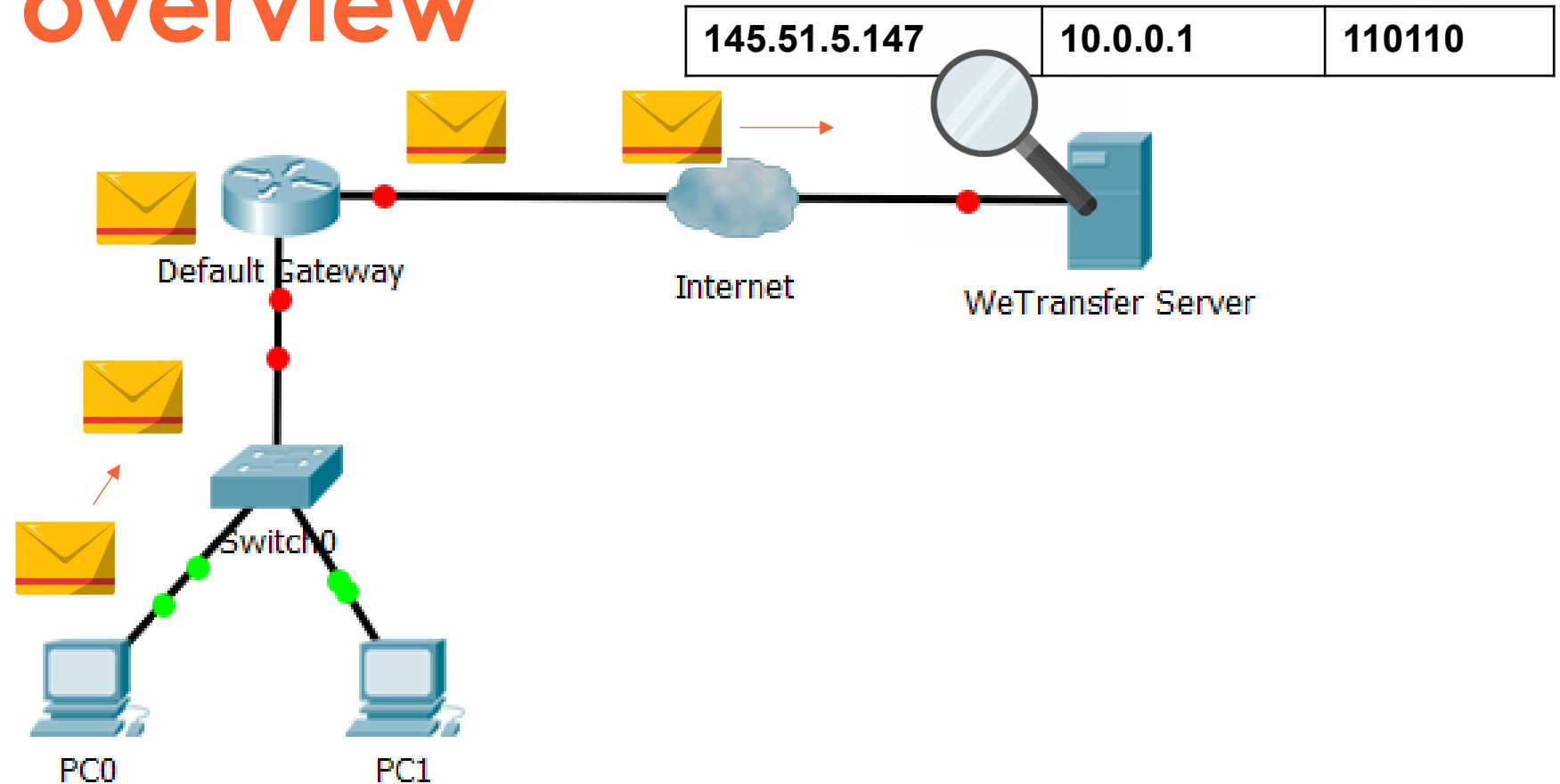


How a packet is send

Very quick overview



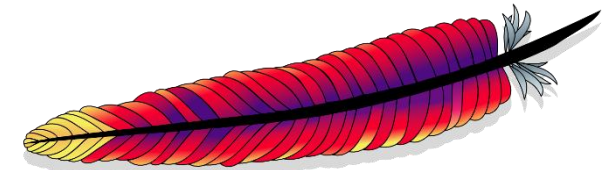
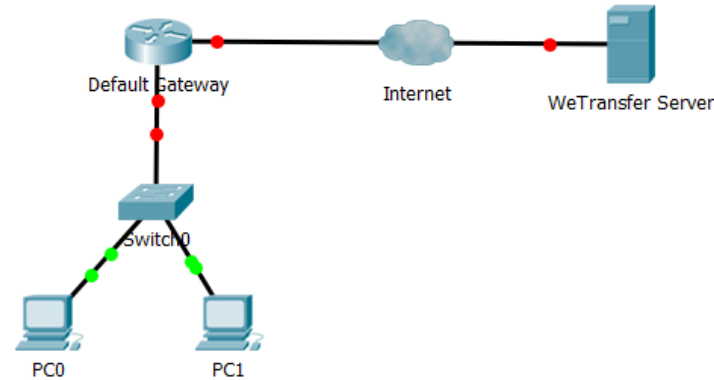
Receiver checks des
Reassembles origina



How a packet is send

Very quick overview

Data is forwarded to the appropriate

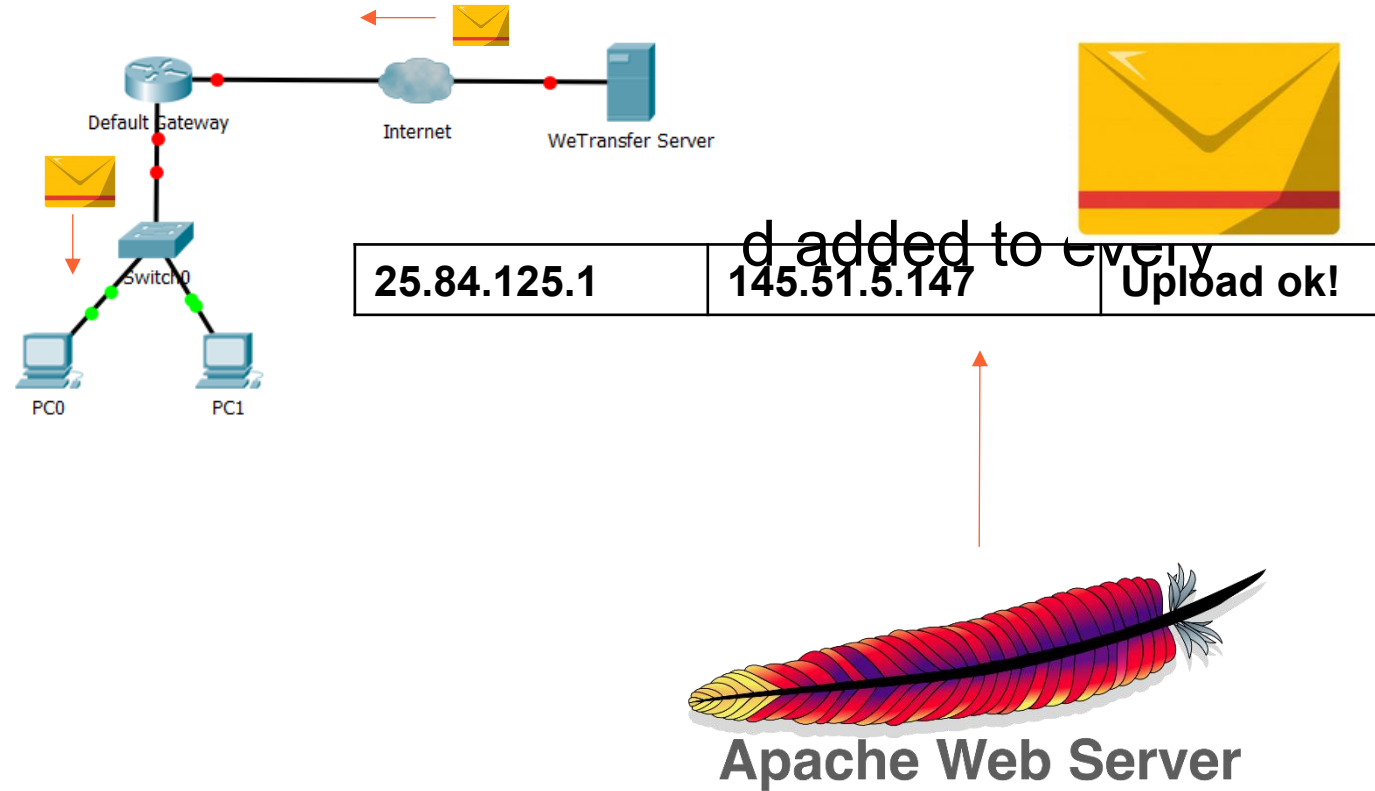


Apache Web Server

How a packet is send

Very quick overview

Application generates response = data
(Split into smaller packets of necessity)
Source address is used as destination
packet



Wireshark packet sniffer

Inspect packets entering and leaving network interface

